

$$\textcircled{1} \frac{\partial f}{\partial x} = 2x + 2y - 1, \quad \frac{\partial f}{\partial y} = 2x + 6y + 2$$

Normální vektor křivky $v(x, y, f(x, y))$

je $(2x + 2y - 1, 2x + 6y + 2, -1)$

Norm. vekt. křivky $2x - 3y + z = 6$ je $(2, -3, 1)$.

Proč $(2x + 2y - 1, 2x + 6y + 2, -1) = \lambda(2, -3, 1)$.

Odtud $\lambda = -1$, a tedy

$$2x + 2y = -1$$

$$2x + 6y = 1$$

$$\Rightarrow y = \frac{1}{2}, \quad x = -1.$$

Hledaný bod je $a = (-1, \frac{1}{2}, f(-1, \frac{1}{2}))$
 $= (-1, \frac{1}{2}, \frac{11}{4})$.

$$b) D_h f(0, 0) = h \cdot (-1, 2) = -h_1 + 2h_2 = 0$$

$$\Rightarrow h_1 = 2h_2$$

$$1 = \|h\|^2 = h_1^2 + h_2^2 = 5h_2^2 \Rightarrow h_2 = \pm \frac{1}{\sqrt{5}}$$

Hledané vektory jsou $\pm \frac{1}{\sqrt{5}}(2, 1)$.

$$\textcircled{2} \quad \frac{\partial f}{\partial x} = 8x - 6x^2 - 2xy = 0 \quad (1)$$

$$\frac{\partial f}{\partial y} = -x^2 + y = 0 \quad (2)$$

$$(2) \Leftrightarrow y = x^2$$

Dosadením do (1) máme $-2x(-4+3x+x^2)=0$

$$\Leftrightarrow x(x+4)(x-1)=0$$

Stacionárne body jsou $a = (0, 0)$, $b = (1, 1)$,
 $c = (-4, 16)$.

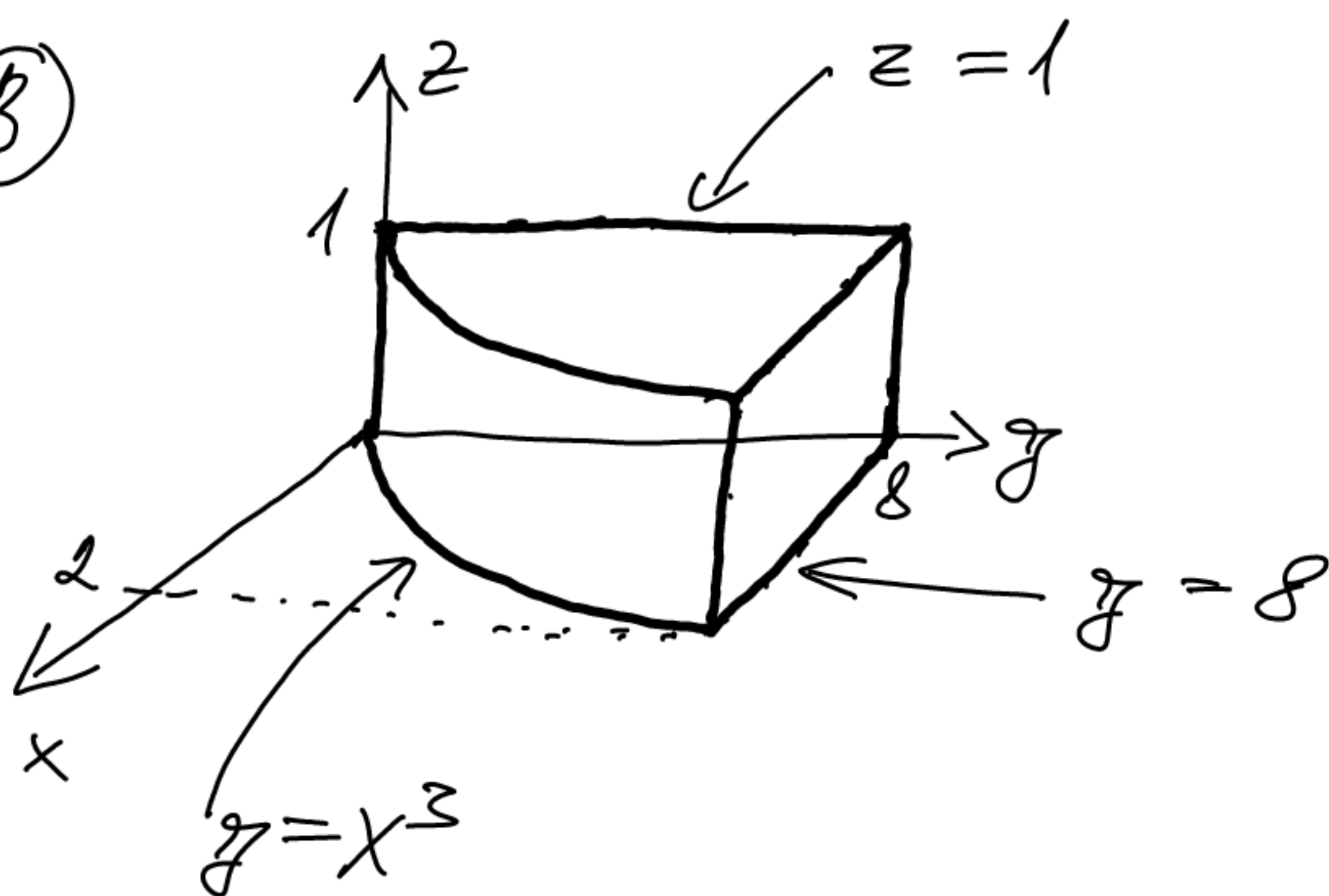
$$H_f(x, y) = \begin{pmatrix} 8 - 12x - 2y & -2x \\ -2x & 1 \end{pmatrix}$$

• $H_f(a) = \begin{pmatrix} 8 & 0 \\ 0 & 1 \end{pmatrix}$... pozit. def. $\Rightarrow a$ je
 bod lok. minima.

• $H_f(b) = \begin{pmatrix} -6 & -2 \\ -2 & 1 \end{pmatrix}$... $\det H_f(b) = -10 < 0 \Rightarrow$
 $\Rightarrow H_f(b)$ je indef. $\Rightarrow b$ je sedlový bod.

• $H_f(c) = \begin{pmatrix} 24 & 8 \\ 8 & 1 \end{pmatrix}$... $\det H_f(c) = 24 - 64 < 0 \Rightarrow$
 $\Rightarrow H_f(c)$ je indef. $\Rightarrow c$ je sedlový bod.

③

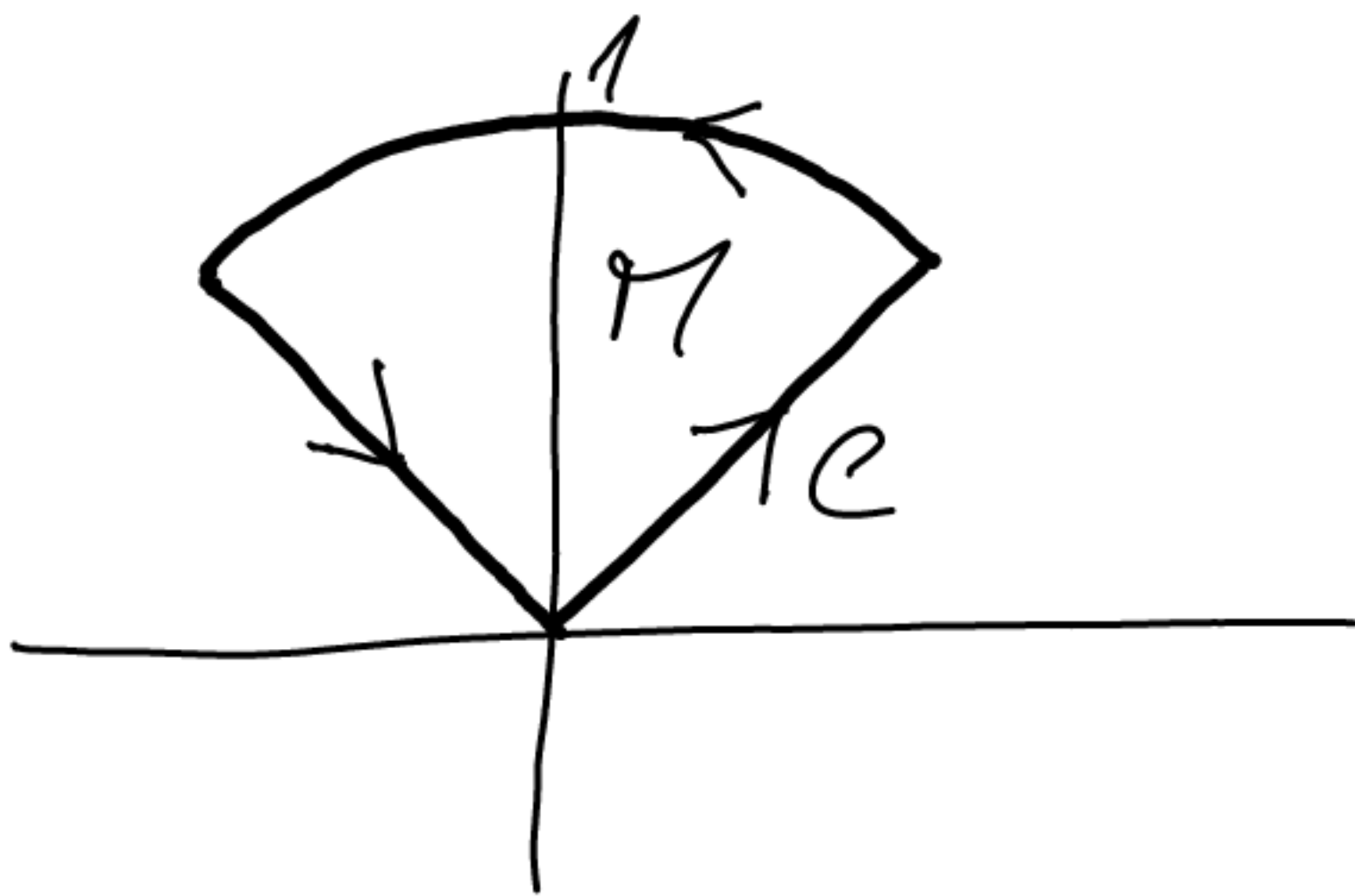


$$\int_H z = \int_0^1 \int_0^2 \int_{x^3}^8 z \, dy \, dx \, dz$$

$$= \int_0^1 \int_0^2 z(8-x^3) \, dx \, dz = \int_0^1 \left(16 - \frac{x^4}{4}\right) z \, dz$$

$$= 12 \int_0^1 z \, dz = 6.$$

④



$$\int_C F = \int_{\pi} \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} d\lambda_2(x, y) = \int_{\pi} 2x + 3y^2 d\lambda_2(x, y)$$

$$= \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \int_0^1 (2x \cos \varphi + 3x^2 \sin^2 \varphi) x dx d\varphi$$

$$= \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{2}{3} \cos \varphi + \frac{3}{4} \sin^2 \varphi d\varphi = \frac{3}{4} \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \sin^2 \varphi d\varphi$$

$$= \frac{3}{8} \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} 1 - \cos 2\varphi d\varphi = \frac{3\pi}{16} - \frac{3}{16} [\sin(2\varphi)]_{\frac{\pi}{4}}^{\frac{3\pi}{4}}$$

$$= \frac{3\pi}{16} - \frac{3}{16} (-1 - 1) = \frac{3}{16} (\pi + 2).$$

$$\begin{aligned}
 \textcircled{5} \sum_{k=0}^{\infty} (k+1)(x-1)^k &= \left(\sum_{k=0}^{\infty} (x-1)^{k+1} \right)' \\
 &= \left[\frac{x-1}{1-(x-1)} \right]' = \left(\frac{x-1}{2-x} \right)' \\
 &= \frac{2-x+x-1}{(2-x)^2} = \frac{1}{(2-x)^2} \quad \text{pro } |x-1| < 1 \\
 &\quad \quad \quad (tj: R=1)
 \end{aligned}$$

Interval konvergence je $(0, 2)$, a ~~pro~~ u bodi -1 řada diverguje.